

Sampling Distribution of a Difference in Means

Shape	Center	Spread
As long as... $n_1 \geq 30$ and $n_2 \geq 30$ to use the _____ if one or both are less than 30, each population should be _____	As long as.... Both samples must have been either: • •	As long as... If sampling, <i>*If this is an experiment, there is no need to check this.</i>

Standard Error

• Because we will rarely know σ, we use the sample standard deviation S_x instead. • This introduces some error, however. • The standard error is also referred to as the estimated standard deviation	
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Degrees of Freedom

Option 1

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1} \left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1} \left(\frac{s_2^2}{n_2}\right)^2}$$

- Note that this option rarely gives degrees of freedom as a whole number (this is what your calculator will give)

Option 2

- Degrees of freedom is _____
- This is considered a “conservative” method
 - Confidence intervals will have a margin of error _____ is needed
 - The significance test gives a p-value _____ the true p-value.

Conditions for Performing Inference

- Random
 - Two random samples (Sampling)
 - Randomly Assigned to two groups (Experiment)
- Independent
 - Sampling – Check 10% condition on BOTH
 - Experiment – Condition doesn’t need to be checked
- Normal/Large Sample
 - Both populations are Normally Distributed
 - Both sample sizes are larger than 30 (CLT)
 - If one or both are smaller than 30, graph to check for no outliers or strong skewness

Two Sample T Interval for a Difference between Two Means

When the conditions are met, an approximate C% confidence interval for $\mu_1 - \mu_2$ is

t^* is the critical value with C% of its area between $-t^*$ and t^* for the t distribution with degrees of freedom using:

- smaller value of $n_1 - 1$ and $n_2 - 1$
- technology to report the “combined” df (must run the test on the calculator to see)

Example: Two Sample T Interval for a Difference in Means

Observational studies suggest that drinking at least 4 cups of coffee a day reduces strokes. This is believed to be because coffee contains a range of plant-based compounds, called polyphenols, thought to promote healthy blood vessels and dampen inflammation. In an experiment, 20 healthy women were assigned at random to drink 3 cups of regular coffee a day or drink no coffee for two weeks. The level of polyphenols in their blood was measured before and after the two-week period. Here are the percent changes (before – after) of the polyphenol levels for the subjects in both groups.

Coffee	3.5	8.1	7.4	2.4	4.0	0.7	4.9	8.4	7.0	5.5
No Coffee	3.1	0.5	-3.8	0.3	4.1	-0.6	2.7	1.9	-3.8	0.1

a) Why will we be creating a two sample T interval for this data and not finding the difference between coffee – no coffee and using a one sample T interval on the difference?

b) Construct and interpret a 90% confidence interval for the difference in mean percent changes in polyphenol levels for the coffee and no coffee treatments.

c) Does the interval suggest that coffee is more effective at producing polyphenols? Explain.